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NPTEL ONLINE CERTIFICATION COURSE

Charging Infrastructure

Lecture-41

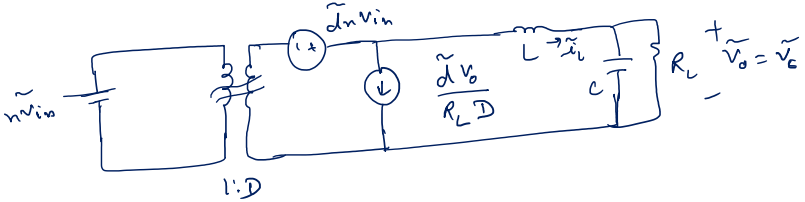
Small Signal Modelling of PSFB-III

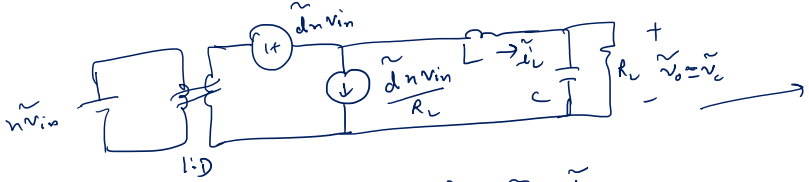
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Recap

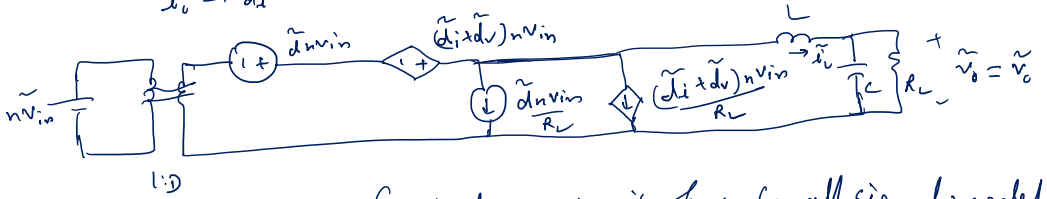




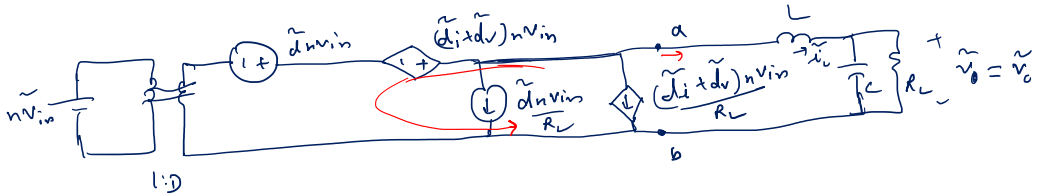
→ Equivalent circuit for small signal model of Buck converter

$$\tilde{d}_{eff_b} = \tilde{d} + \tilde{d}_i + \tilde{d}_v$$

$$\tilde{i}_L \rightarrow \tilde{d}_i \quad \Rightarrow \quad \tilde{v}_{in} \rightarrow \tilde{d}_v$$



⇒ Equivalent circuit for small signal model of PSFB converter



$$\left.G_{vd}(s)\right| = \frac{\tilde{v}_o(s)}{\tilde{d}(s)}$$

$\tilde{v}_{in} = 0$

$\Rightarrow \tilde{d}_v = 0$

$\Rightarrow$

$$\tilde{v}_{ab} = \tilde{d}_v n v_{in} + \tilde{d}_i n v_{in}$$

$$\tilde{v}_{ab} = n v_{in} (\tilde{d} + \tilde{d}_i)$$

$$\Rightarrow \tilde{v}_{ab}(s) = n v_{in} (\tilde{d}(s) + \tilde{d}_i(s)) \longrightarrow (1)$$

$$H_o(s) = \frac{\tilde{v}_o(s)}{\tilde{v}_{ab}(s)}$$

$$\Rightarrow \tilde{v}_o(s) = H_o(s) \cdot \tilde{v}_{ab}(s)$$

$$\Rightarrow \tilde{v}_o(s) = H_o(s)\, n v_{in} (\tilde{d}(s) + \tilde{d}_i(s)) \longrightarrow (2)$$

Input impedance from terminal a & b, (in s-domain)

$$Z_{in(ab)}(s) = sL + \frac{\frac{1}{sC} \cdot R_L}{\frac{1}{sC} + R_L}$$

$$Z_{in(ab)}(s) = \frac{s^2 LCR_L + sL + R_L}{1 + sCR_L} \longrightarrow (3)$$

$$\Rightarrow \tilde{v}_o(s) = \frac{\tilde{v}_{ab}(s)}{Z_{in(ab)}(s)} \times \frac{\frac{1}{sC}}{\left(\frac{1}{sC} + R_L\right)} \times R_L$$

by the current division between $\frac{1}{sC}$ & R_L

current flowing through R_L

$$\Rightarrow \frac{\tilde{v}_o(s)}{\tilde{v}_{ab}(s)} = \frac{(1+sCR_L)}{s^2 LCR_L + sL + R_L} \times \frac{R_L}{(1+sCR_L)}$$

$$\Rightarrow H_o(s) = \frac{1}{s^2 LC + s\frac{L}{R_L} + 1} \longrightarrow (4)$$

Source: J. A. Sabate, V. Vlatkovic, R. B. Ridley, F. C. Lee and B. H. Cho, "Design considerations for high-voltage high-power full-bridge zero-voltage-switched PWM converter," *Fifth Annual Proceedings on APEC*, Los Angeles, CA, USA, 1990, pp. 275-284, doi: 10.1109/APEC.1990.66420.

$$\tilde{d}_i = \frac{-\Delta t}{T_{s/2}}$$

$$\tilde{d}_i = -\frac{4n L_{mg} f_{sw}}{V_{in}} \cdot \tilde{i}_L \longrightarrow (5)$$

$$R_{eq} = \frac{4n^2 L_{mg} f_{sw}}{} \longrightarrow (6)$$

$$\Rightarrow \tilde{i}_L = -\frac{n V_{in}}{R_{eq}} \cdot \tilde{d}_i$$

(from (5) & (6)) $\Rightarrow$ $\tilde{i}_L(s) = -\frac{n V_{in}}{R_{eq}} \cdot \tilde{d}_i(s) \longrightarrow (7)$

$$Z_{in(ol)}(s) = \frac{s^2 LC R_L + sL + R_L}{1 + sCR_L}$$ (from eq. (3))

$$\tilde{i}_L(s) = \frac{\tilde{v}_{ol}(s)}{Z_{in(ol)}(s)}$$

from eq. (1) $\Rightarrow$ $\tilde{i}_L(s) = \frac{n V_{in}\left(\tilde{d}(s) + \tilde{d}_i(s)\right)}{Z_{in,(ol)}(s)} \longrightarrow (8)$

Source: J. A. Sabate, V. Vlatkovic, R. B. Ridley, F. C. Lee and B. H. Cho, "Design considerations for high-voltage high-power full-bridge zero-voltage-switched PWM converter," *Fifth Annual Proceedings on APEC*, Los Angeles, CA, USA, 1990, pp. 275-284, doi: 10.1109/APEC.1990.66420.

from eq (7) & (8)

$$\Rightarrow -\frac{n v_{in}\,\tilde{d}_i(s)}{R_{eq}} = \frac{n v_{in}\left(\tilde{d}(s)+\tilde{d}_i(s)\right)}{Z_{in,ab}(s)}$$

$$\Rightarrow \tilde{d}_i(s) = \left[\frac{-\tilde{d}(s)}{1+\frac{Z_{in,conv}(s)}{R_{eq}}}\right] \rightarrow (9)$$

from eq (2)

$$\tilde{v}_o(s) = H_o(s)\cdot n v_{in}\left(\tilde{d}(s)+\tilde{d}_i(s)\right)$$

$$= H_o(s)\, n v_{in}\left[\tilde{d}(s) - \frac{\tilde{d}(s)}{1+\frac{Z_{in,ab}(s)}{R_{eq}}}\right] \quad (\text{from eq } (9))$$

$$\Rightarrow \tilde{v}_o(s) = H_o(s)\, n v_{in}\left[\frac{Z_{in,ab}(s)}{Z_{in,ab}(s)+R_{eq}}\right] \times \tilde{d}(s)$$

Source: J. A. Sabate, V. Vlatkovic, R. B. Ridley, F. C. Lee and B. H. Cho, "Design considerations for high-voltage high-power full-bridge zero-voltage-switched PWM converter," *Fifth Annual Proceedings on APEC*, Los Angeles, CA, USA, 1990, pp. 275-284, doi: 10.1109/APEC.1990.66420.

$$\rightarrow G_{vd}(s) = \frac{\tilde{v}_o(s)}{\tilde{d}(s)} = H_o(s) \times V_{in} \left[\frac{Z_{in(ab)}(s)}{Z_{in(cap)}(s) + R_{eq}} \right]$$

$Z_{in,(ab)}(s)$ from eq. (3), $H_o(s)$ from eq. (4)

$$G_{vd}(s) = \frac{1}{s^2 LC + sL/R_L + 1} \cdot n \cdot V_{in} \left[\frac{\frac{s^2 LCR_L + sL + R_L}{(1+sCR_L)}}{\frac{s^2 LCR_L + sL + R_L}{(1+sCR_L)} + R_{eq}} \right]$$

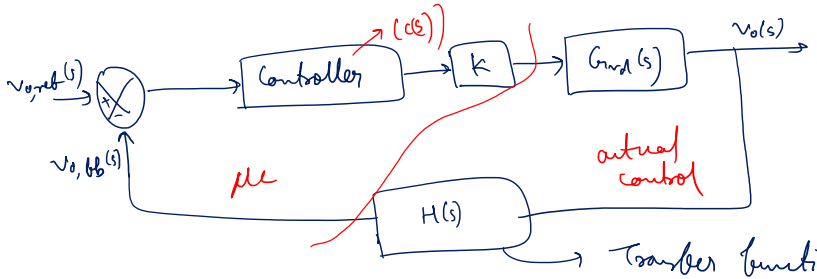
$$= \frac{R_L}{(s^2 LCR_L + sL + R_L)} \cdot n \cdot V_{in} \left[\frac{(s^2 LCR_L + sL + R_L)}{s^2 LCR_L + s(L + CR_L R_{eq}) + R_{eq}} \right]$$

$$G_{vd}(s)\Big|_{\tilde{v}_{in}=0} = \frac{nV_{in}}{s^2 LC + s\left(\frac{L}{R_L} + CR_{eq}\right) + \frac{R_{eq}}{R_L}}$$

where $\left(R_{eq} = 4n^2 L_{leg}\, f_{sw}\right)$

↳ Transfer function of PSFB which captures the output voltage dynamics whenever there is a change in the duty ratio

Source: J. A. Sabate, V. Vlatkovic, R. B. Ridley, F. C. Lee and B. H. Cho, "Design considerations for high-voltage high-power full-bridge zero-voltage-switched PWM converter," *Fifth Annual Proceedings on APEC*, Los Angeles, CA, USA, 1990, pp. 275-284, doi: 10.1109/APEC.1990.66420.



$$OLTF(s) = C(s) \cdot K \cdot G_{vd}(s) \cdot H(s) \quad \rightarrow (10)$$

→ Transfer function of output voltage sensor

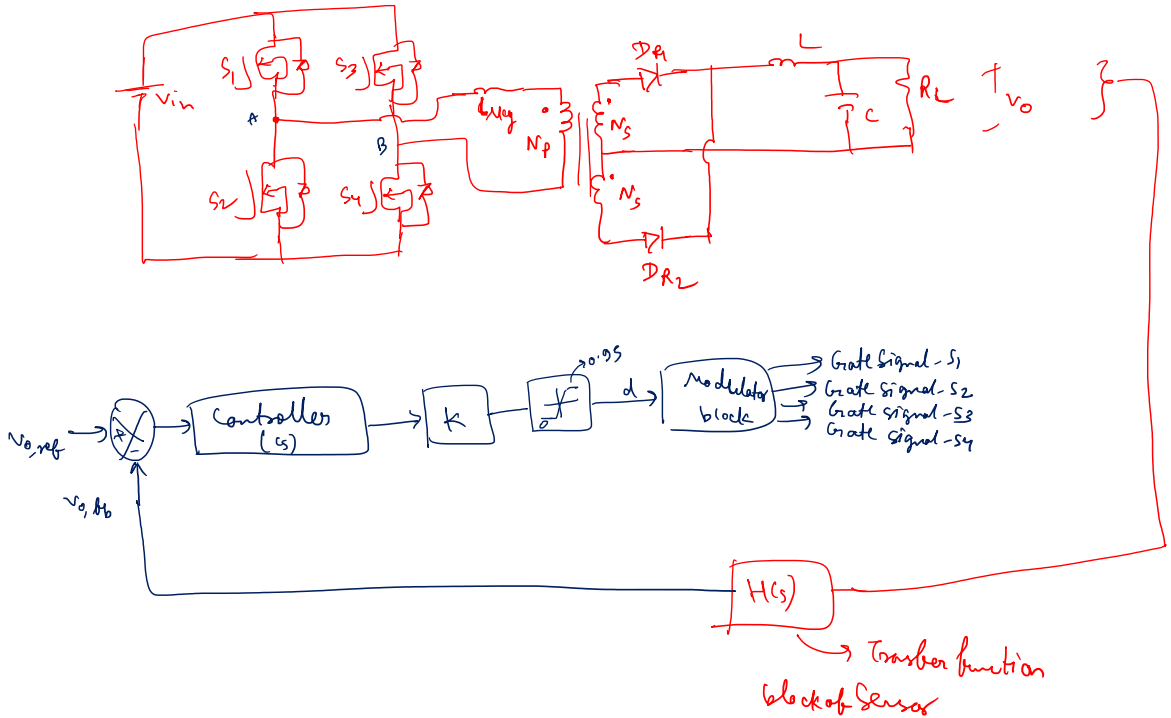
The K is the gain $\Rightarrow K = \dfrac{\text{Max. control voltage } (V_c)}{\text{Max. output voltage}}$

⇒ The guideline for Controller Parameters

→ Provide high gain at the freq. (which the closed loop system need to track down) (in our case the high gain needs to be provided at $s=0$)

→ Sufficient PM for $OLTF(s)$ (from eq (10)), this can be done using bode plots ($PM > 35°$)

→ Sufficient Bandwidth to be provided such that the controller doesn't get overburden and closed loop system is able to track down frequency of concern.


V_{in}
S_1
S_2
S_3
S_4
A
B
N_p
N_s
N_s
D_{R1}
D_{R2}
L
C
R_L
V_o
$V_{o,ref}$
Controller (G)
K
0.95
d
Modulator block
Gate Signal - S_1
Gate Signal - S_2
Gate signal - S_3
Gate signal - S_4
$V_{o,fb}$
H(s)
Transfer function block of Sensor
d
Modulator block
PS signal

Thank You